

Mock Test 01

Class X – Coordinate Geometry

Time Allowed: 40 Minutes

Maximum Marks: 20

General Instructions:

- All questions are compulsory.
- The question paper comprises four sections: **A, B, C,** and **D.**
- Section A contains Multiple Choice and Assertion-Reason questions of 1 mark each.
- Section B contains Short Answer questions of 2 marks.
- Section C contains Long Answer questions of 5 marks.
- Section D contains 1 Case-based integrated question of 4 marks.

SECTION – A

(1 Mark Each)

1. Identify the false statement among the following:
(a) Every point on the x -axis has y -coordinate 0.
(b) The origin lies on both the axes.
(c) $(2, 3)$ and $(3, 2)$ are the same point.
(d) Points in Quadrant II have negative x -coordinate and positive y -coordinate.
2. The point which lies on the y -axis at a distance of 5 units in the negative direction of the y -axis is:
(a) $(5, 0)$ (b) $(0, 5)$ (c) $(-5, 0)$ (d) $(0, -5)$
3. If the distance between the points $A(2, -2)$ and $B(-1, x)$ is equal to 5, then the value of x is:
(a) 2 (b) -2 (c) 1 (d) -1
4. If $O\left(\frac{p}{3}, 4\right)$ is the midpoint of the line segment joining the points $P(-6, 5)$ and $Q(-2, 3)$, then the value of p is:
(a) $7/2$ (b) -12 (c) 4 (d) -4
5. What is the slope of the line $y = 5x - 3$?
(a) -3 (b) 5 (c) -5 (d) 2
6. The point $(3, 4)$ lies on the graph of the equation $3y = kx + 7$. The value of k is:
(a) $4/3$ (b) $5/3$ (c) 3 (d) $7/3$
7. **Direction:** In this question, a statement of Assertion (A) is followed by a statement of Reason (R).
Assertion (A): Point $A(-7, -9)$ lies in III quadrant.
Reason (R): A point both of whose coordinates are negative lies in III quadrant.
(a) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion.

- (b) Both Assertion and Reason are correct but Reason is not the correct explanation for Assertion.
- (c) Assertion is true but Reason is false.
- (d) Assertion is false but Reason is true.

SECTION – B

8. Find the value of y if the distance between the points $(3, -2)$ and $(8, y)$ is $\sqrt{89}$ units. **[2 Marks]**
9. Verify whether the points $A(0, 0)$, $B(3, 4)$, and $C(7, 4)$ form a right-angled triangle. Identify the right angle. **[2 Marks]**

SECTION – C

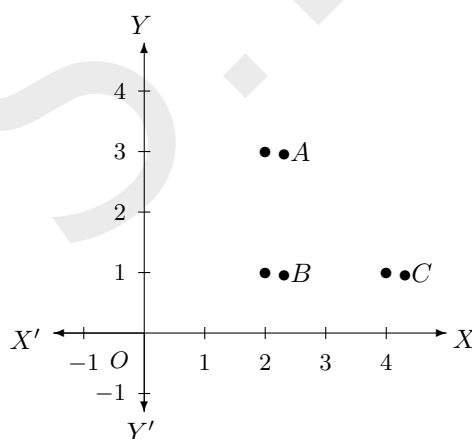
10. Prove that the points $A(1, 7)$, $B(4, 2)$, $C(-1, -1)$, and $D(-4, 4)$ are the vertices of a square. Show all four side-lengths and both diagonal lengths, and explain why the conditions taken together imply that $ABCD$ is a square (and not merely a rhombus or rectangle). **[5 Marks]**

SECTION – D

11. Case Study:

[1 + 1 + 2 = 4 Marks]

Alia and Shagun are friends living on the same street in Patel Nagar. Shagun's house is at the intersection of one street with another street on which there is a library. They both study in the same school and that is not far from Shagun's house. Suppose the school is situated at the point O , i.e., the origin, Alia's house is at A , Shagun's house is at B and library is at C .



Based on the above information, answer the following questions:

- (a) How far is Alia's house from Shagun's house? **(1 Mark)**
- (b) How far is the library from Shagun's house? **(1 Mark)**
- (c) Show that for Shagun, school is farther compared to Alia's house and library. **(2 Marks)**

OR

- (d) Show that Alia's house, Shagun's house, and the library form an isosceles right triangle. **(2 Marks)**